Neel Gali

AI/ML Engineer

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Professional Experience

Software Engineer AI / ML *Jun 2025 - Present*

Epik Solutions

- Fine-tuned transformer-based NLP models (BERT, RoBERTa) in PyTorch for mobile-optimized search features, improving query accuracy by 32% and reducing p95 inference latency to under 80ms on-device.
- Applied model quantization and structured pruning via TensorFlow Lite, shrinking deployed model size by 40% with under 2% accuracy regression, enabling direct on-device inference for mobile app features.
- Designed end-to-end ML pipelines covering feature engineering, distributed training, A/B evaluation, and drift monitoring within 24 hours of deployment., cutting model retraining cycle time by 45%
- Applied gradient boosting and statistical modeling across 10M+ user interaction events to surface feature importance signals, driving data-informed product decisions that lifted session engagement by 18%.
- Optimized Golang microservices for real-time model serving, offloading 60% of prediction calls to on-device TensorFlow Lite models, eliminating server round-trips and average inference latency by 15%.

Software Engineer (AI/Search) *Jul 2024 - Jun 2025*

Samay Systems LLC

- Built and deployed production-grade ranking models on AWS SageMaker for a multi-tenant search system serving 15+ clients, maintaining 99.9% uptime and sub-100ms response at peak load.
- Implemented an LLM-powered NLP pipeline (HuggingFace Transformers + LangChain) for automated compliance document summarization, processing 10K+ documents monthly and cutting effort by 70%.
- Developed computer vision preprocessing modules combining OCR and layout parsing for document ingestion, improving structured data extraction F1-score from 0.71 to 0.89 across 500K+ artifacts.
- Architected a real-time ML feature store on AWS (Kinesis + S3 + SageMaker Feature Store), enabling continuous model retraining on fresh data and reducing drift accuracy degradation by 35%.
- Instrumented model monitoring dashboards tracking prediction drift, data quality, and latency SLAs across 10+ deployed models, enabling automated retraining triggers and >92% accuracy in production.

SDE Intern *Jun 2023 - Aug 2023*

Monarch Tractor

- Built Python MLOps infrastructure using active learning strategies for data labeling, reducing annotation cost by 35% while boosting throughput by 20% across 500K+ agricultural imagery samples in training
- Deployed serverless AWS Lambda pipelines for version-controlled dataset delivery to ML training jobs, cutting pipeline failures by 40% and sustaining continuous training across 3 production vision models.

Software Engineer *Jul 2021 - Aug 2022*

ZeMoSo Technologies

- Integrated automated anomaly detection modules using statistical modeling (z-score, IQR-based flagging) into the lab platform's data layer, reducing manual review time by 30% for 500+ scientists.
- Designed features across 10+ product modules using React, Ruby on Rails, Spring Boot, and PostgreSQL in a microservices architecture for scalability.
- Authored 100+ test cases and resolved critical production issues, enhancing system reliability and data integrity in a regulated scientific platform.

Skills

Programming Languages: Python, Golang, C#, Java, TypeScript, SQL, Bash
Machine Learning & AI: Industrial Search Recommendation, NLP Algorithms, Large Model Applications, Machine Learning Theory, Search Algorithm Models, NLP Algorithm Models, Deep Learning Frameworks, TensorFlow, PyTorch, RAG, LangChain
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Terraform, Git, CI/CD
Frameworks & Libraries: React, GraphQL, Node.js, ASP .NET Core, Spring Boot, Gin
Databases & Data: PostgreSQL, MySQL, MongoDB, Redis, OpenSearch, Neo4j, Data Pipelines

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India